



Basic Principles of Medicine 2

Module: Endocrine and Reproductive System | DLA

Autonomic Innervation of the Male Reproductive and Pelvic Organs

Department of Anatomical Sciences

School of Medicine, St. George's University

Copyright

All year-1 course materials, whether in print or online, are protected by copyright. The work, or parts of it, may not be copied, distributed, or published in any form, printed, electronic, or otherwise.

As an exception, students enrolled in year 1 of St. George's University School of Medicine and their faculty are permitted to make electronic or print copies of all downloadable files for personal and classroom use only, provided that no alterations to the documents are made and that the copyright statement is maintained in all copies.

'View only' files, such as lecture recordings, are explicitly excluded from download, and creating copies of these recordings by students and other users is strictly illegal.

The author of this document has made the best effort to observe current copyright law and the copyright policy of St. George's University. Users of this document identifying potential violations of these regulations are asked to bring their concerns to the attention of the author.

Objectives

SOM.MKII.BPM2.1.ER.1.ANAT.1049	Describe the origin, and type of nerve fibers, found in the superior hypogastric plexus and the inferior hypogastric (pelvic) plexus
SOM.MKII.BPM2.1.ER.1.ANAT.1114	Describe the autonomic innervation of the male pelvic organs and the relationship to the pelvic pain line
SOM.MKII.BPM2.1.ER.1.ANAT.1038	Compare and contrast the visceral pain pathways for structures above the peritoneal floor of the pelvis versus structures below the peritoneal floor (pelvic pain line) in males



Autonomic functions of the pelvic organs

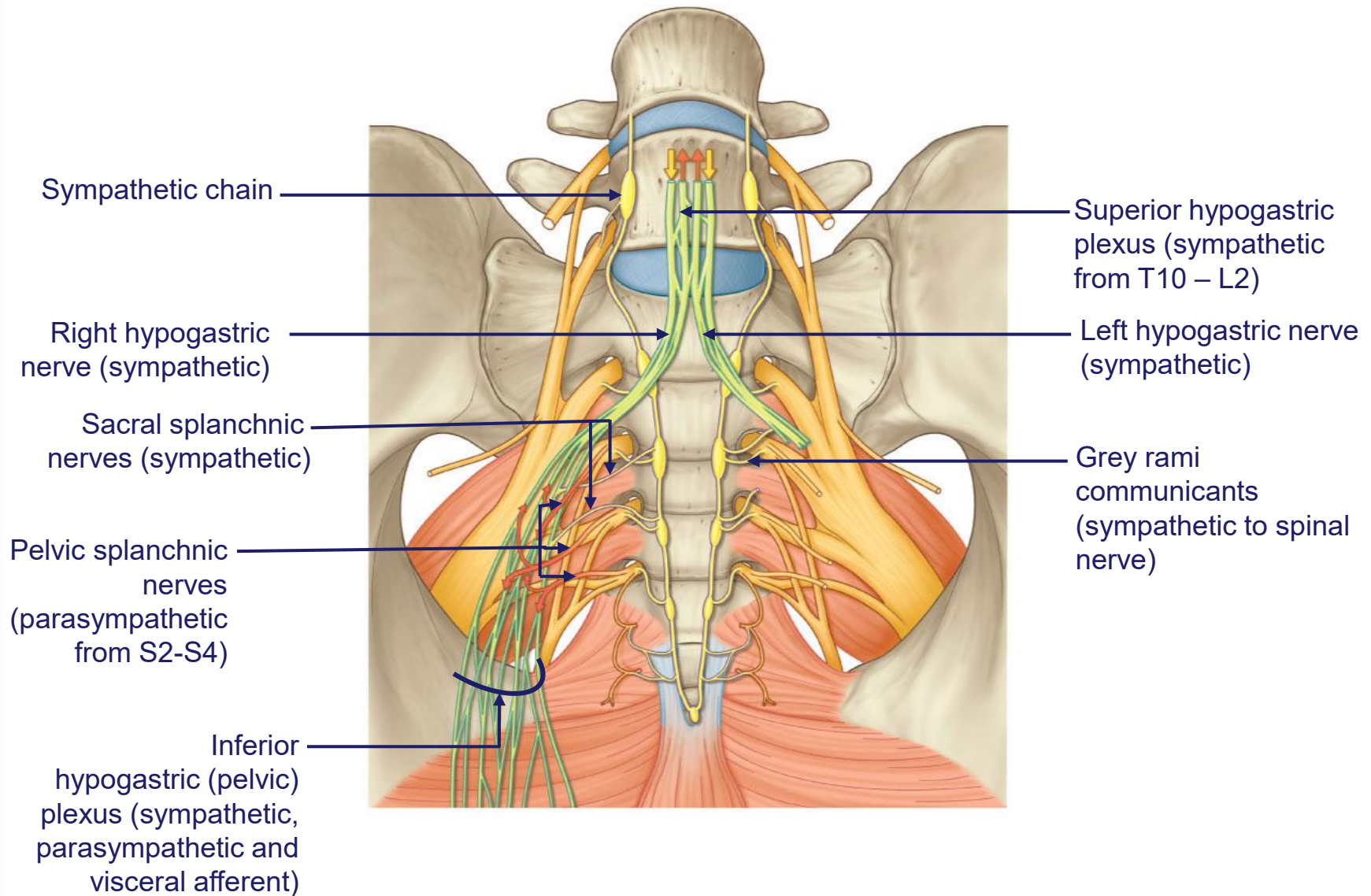
Sympathetic functions:

- Contraction of smooth muscle in blood vessels (vasoconstriction)
- Smooth muscle contraction in seminal vesicle, prostate and bulbourethral glands during emission. This helps move secretions into the urethra
- Sphincter constriction - internal urethral and internal anal maintaining continence
- Relaxation of detrusor muscle of the bladder to promotes bladder filling

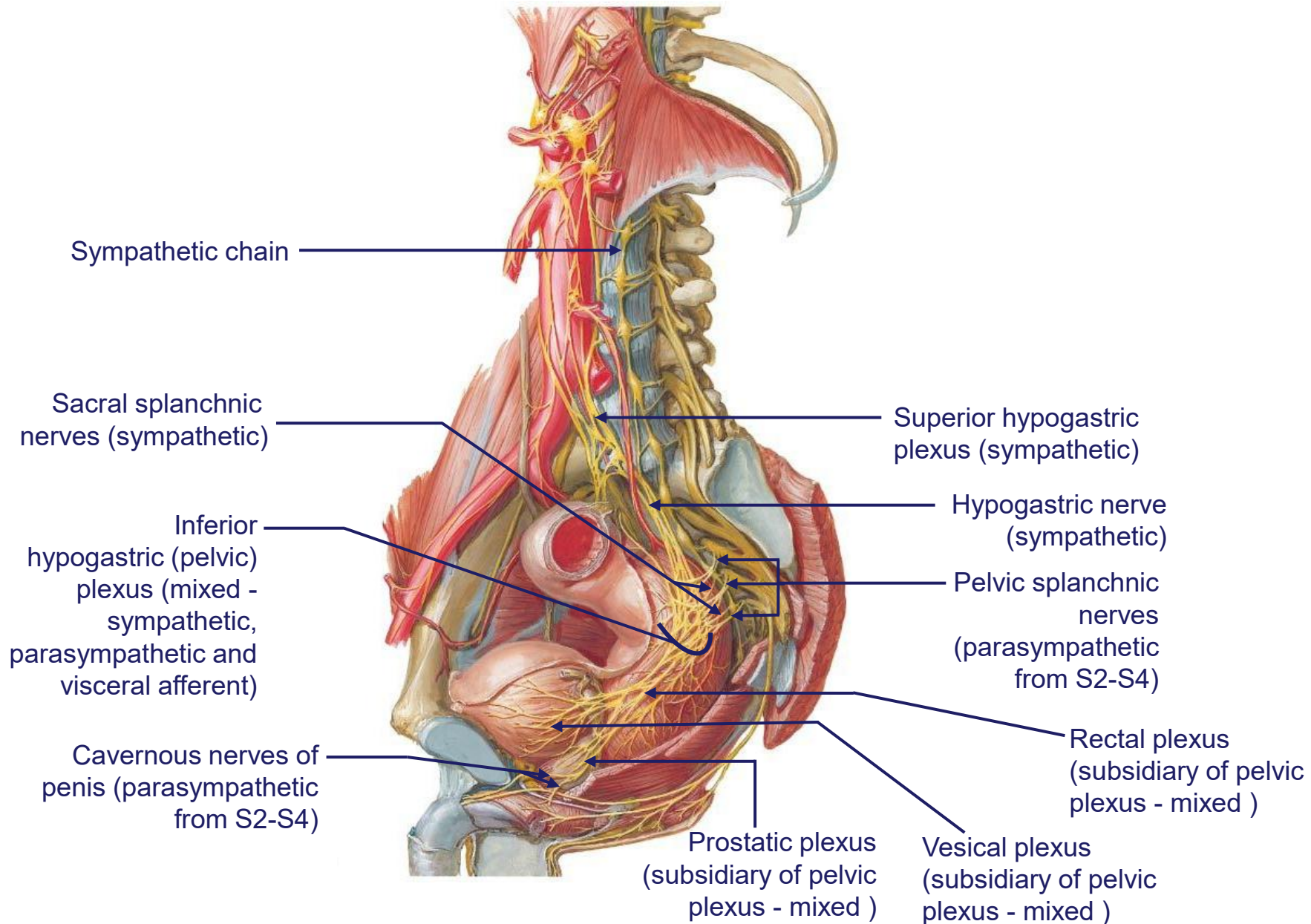
Parasympathetic functions:

- Elicits vasodilation of erectile tissues in the penis, leading to erection (via the cavernous nerves).
- Stimulates glandular secretions
- Stimulates contraction of the detrusor muscle, promoting micturition
- Sphincter relaxation - internal urethral and internal anal – promoting micturition and defecation
- Stimulates peristalsis in the rectum, aiding in defecation

Autonomic innervation of pelvic organs



Autonomic innervation of male pelvic organs



Autonomic innervation of pelvic organs

- The autonomic nerves of the pelvis form a complex network that is primarily responsible for the regulation of the pelvic organs, including the bladder, rectum, and sexual organs
- These nerves include both sympathetic and parasympathetic fibers, which often have opposing actions

- 1. Superior Hypogastric Plexus (Predominantly Sympathetic):** This plexus is located anterior to the lower part of the abdominal aorta and bifurcation. It gives rise to the hypogastric nerves
- 2. Hypogastric Nerves (Sympathetic):** These nerves descend from the superior hypogastric plexus along the sides of the rectum to join the pelvic plexuses

Autonomic innervation of pelvic organs

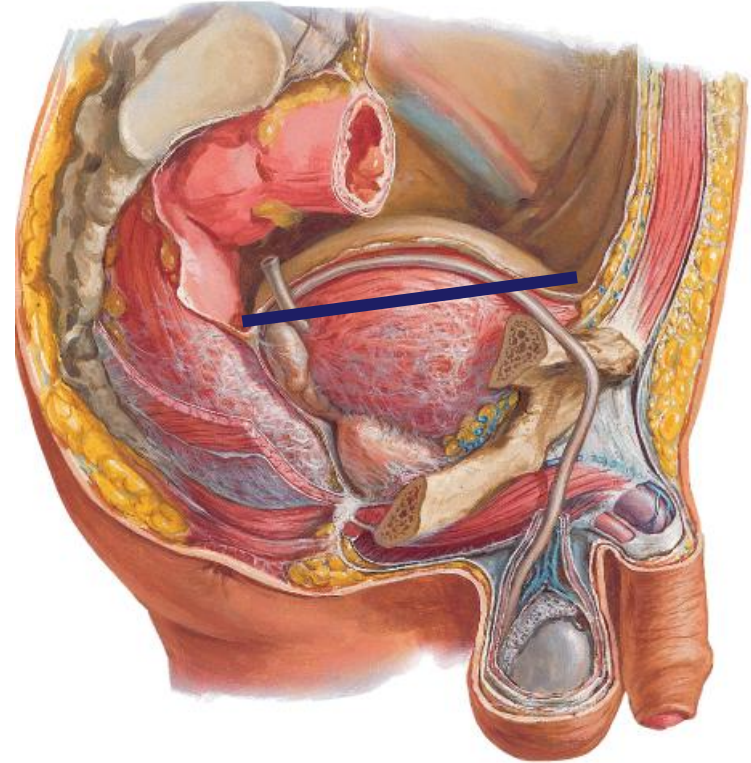
3. Pelvic Splanchnic Nerves (Parasympathetic): These nerves arise from the S2-S4 spinal segments and carry parasympathetic fibers to the pelvic organs.

4. Sacral Splanchnic Nerves (T10-L2) (Sympathetic): These nerves arise from the sacral part of the sympathetic trunk, and they carry sympathetic fibers to the pelvic organs.

5. Inferior Hypogastric Plexus (Mixed Sympathetic and Parasympathetic): Also known as the pelvic plexus, it lies on the side wall of the pelvis, in front of the piriformis muscle and at the sides of the rectum. **It receives the hypogastric nerves, the sacral splanchnic nerves (sympathetic) and pelvic splanchnic nerves (parasympathetic).** Subsidiary plexuses from this plexus innervate the bladder, rectum, sex organs, and other pelvic structures.

Pelvic pain line (PPL)

- Imaginary line corresponds to the lower limit of the peritoneum
- **Above PPL (peritoneal portion of bladder)** - the visceral afferents for pain travel via the **sympathetics**
 - They travel in the inferior hypogastric plexus
 - **Pain is referred to the T10-L2 areas**
- **Below PPL (majority of the urinary bladder, prostate, seminal vesicle)** - the visceral afferents for pain travel via the **parasympathetics**
 - They travel in the inferior hypogastric plexus
 - **Pain is referred to the S2-S4 areas**



Pelvic pain line

- The concept of the "pelvic pain line" is often used in anatomy and medicine to understand the patterns of pain sensation in the pelvis. It's essentially a boundary within the pelvis that separates areas with different types of sensory innervation:
 - **Above the Pelvic Pain Line (peritoneal portion of bladder):** GVA fibers from structures above the pain line follow the course of the sympathetic efferent fibers. **Pain is referred to the T10-L2 areas**
 - **Below the Pelvic Pain Line (majority of the urinary bladder, prostate, seminal vesicle):** GVA fibers from structures below the pain line follow the course of the parasympathetic efferent. **Pain is referred to the S2-S4 areas**